REVISION WORKSHEET

UNITED ARAB EMIRATES
MINISTRY OF EDUCATION

الإمارات العربية المتحدة
وزارة التربية والتعليم

AL KHALIDIYA BOYS SCHOOL

Cycle 3 (10048) | Dibba Al Hisn | Sharjah Eastern

Where and When Practice Questions

Grade 9 Advanced | Module 2 Lesson 2 | Where and When

Name: _____ Class: _____ Date: _____

Answer all 30 questions. Select one option for each MCQ. Show equations, working, units and explanations for written questions.

Useful equations and guidance: $\Delta x = x_f - x_i$; $\Delta t = t_f - t_i$; distance adds path lengths; choose an origin and positive direction.

Term 1 2026–2027 | Textbook alignment: uploaded Grade 9 book, printed pages 32–36.

P01. Displacement is:

A. Total path length only B. Always positive
C. Change in position D. A scalar without direction

P02. Which is a vector quantity?

A. Mass B. Time
C. Distance D. Displacement

P03. A coordinate origin specifies:

A. The chosen zero position B. The direction of gravity everywhere
C. The object's speed D. The end of every trip

P04. A negative position means the object is:

A. Necessarily moving left B. On the negative side of the chosen origin
C. Necessarily slowing down D. At a negative distance travelled

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P05. An object returns to its starting point. Its net displacement is:

A. Equal to total distance B. Always positive
C. Zero D. Undefined

P06. An object starts at x = -4 m and moves 10 m in the positive direction. Calculate its final position.

A. 6 m B. 12 m
C. 3 m D. -6 m

P07. An object moves from x = 6 m to x = -4 m. Calculate its displacement.

A. -20 m B. -10 m
C. -5 m D. 10 m

P08. A student walks 10 m east and then 4 m west. Taking east as positive, calculate the displacement.

A. 12 m B. 3 m
C. 6 m D. -6 m

P09. An observation starts at t = 4 s and ends at t = 14 s. Calculate the time interval.

A. 20 s B. 5 s
C. -10 s D. 10 s

P10. A student walks 10 m east and then 4 m west. Calculate the total distance travelled.

A. 14 m B. 28 m
C. 7 m D. -14 m

P11. Define a scalar quantity.

P12. Define a vector quantity.

P13. Distinguish distance from displacement.

P14. What choices define a one-dimensional coordinate system?

P15. Does changing the origin change the displacement of a fixed trip?

P16. An object starts at x = -6 m and moves 15 m in the positive direction. Calculate its final position.

P17. An object starts at x = -8 m and moves 20 m in the positive direction. Calculate its final position.

P18. An object moves from x = 9 m to x = -6 m. Calculate its displacement.

P19. An object moves from x = 12 m to x = -8 m. Calculate its displacement.

P20. A student walks 15 m east and then 6 m west. Taking east as positive, calculate the displacement.

P21. A student walks 20 m east and then 8 m west. Taking east as positive, calculate the displacement.

P22. An observation starts at t = 6 s and ends at t = 21 s. Calculate the time interval.

P23. An observation starts at t = 8 s and ends at t = 28 s. Calculate the time interval.

P24. A student walks 15 m east and then 6 m west. Calculate the total distance travelled.

P25. A student walks 20 m east and then 8 m west. Calculate the total distance travelled.

P26. A student walks from x = −4 m to x = +6 m and then to x = +1 m. Find distance and displacement.

P27. Give an example of nonzero distance with zero displacement.

P28. Can a negative position be combined with a positive displacement? Explain with numbers.

P29. A student adds the magnitudes of two opposite displacements to find the net displacement. Evaluate the method.

P30. Redraw a trip after moving the coordinate origin 10 m right. Explain what remains unchanged.